

ADVANCED JAVA LABSIDDHARTHA ACADEMY OF HIGHER EDUCATION (DEEMED TO BE UNIVERSITY)
(23AI5351) • DEPARTMENT OF AI • BATCH 2025–2029

Week 02: Control Flow Structures & 2D Array Operations

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1. OBJECTIVE / AIM

To implement control statements (conditional branching and loops) and perform 2D matrix multiplication and transposition in Java.

2. CONCEPT & THEORY

Control structures regulate the flow of program execution based on conditional expressions. In Java, arrays are dynamically allocated objects that store a fixed number of elements of the same type.

3. JAVA SOURCE IMPLEMENTATION

```
// Advanced Java Lab - Experiment 02
// Student: Trivikram | Roll No: 25EU02067
public class MatrixMultiplication {
    public static void main(String[] args) {
        int[][] a = {{1, 2}, {3, 4}};
        int[][] b = {{5, 6}, {7, 8}};
        int[][] c = new int[2][2];

        System.out.println("Matrix Multiplication (A x B):");
        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                c[i][j] = 0;
                for (int k = 0; k < 2; k++) {
                    c[i][j] += a[i][k] * b[k][j];
                }
                System.out.print(c[i][j] + "\t");
            }
            System.out.println();
        }
    }
}
```

4. COMPILATION & EXECUTION OUTPUT

Matrix Multiplication (A x B):

19 22

43 50